

Janith Kavindu Dassanayake

PH.D. CANDIDATE (ABD) · SCHOOL OF ELECTRICAL, COMPUTER, AND BIOMEDICAL ENGINEERING

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Education

Southern Illinois University Carbondale

Carbondale, IL

PHD ELECTRICAL AND COMPUTER ENGINEERING

Jan. 2022 - Present

- Current GPA: 4.00
- Advisor: Dr. Gayan Aruma Baduge
- Dissertation Proposal: *"Waveform and Resource Optimization for Wireless Communications, Sensing, and Energy Transfer"*
Content: Massive multiple-input-multiple-output (MIMO), Integrated Sensing and Communication (ISAC), Near-field Communication, Extremely Large Aperture Arrays (ELAAs), Simultaneous Wireless Information and Power Transfer (SWIPT), Reconfigurable intelligent surfaces (RISs), Physical Layer Security, Convex Optimization, Deep learning for wireless communication

University of Ruhuna

Hapugala, Galle, Sri Lanka

BSc (HONS) ELECTRICAL AND INFORMATION ENGINEERING

Jan. 2016 - Mar. 2020

- GPA: 3.62 (Second Class Upper Division Honors)

Research Interests

- Optimal Waveform and beamformer designs for ISAC systems
- Tri-hybrid precoder optimization for flexible ultra MIMO systems
- Distributed/cell-free massive MIMO architectures
- Reconfigurable active/passive intelligent surfaces
- Learning-based optimization and signal processing

Areas of Expertise

- Wireless Communications: Channel modeling and estimation
- Deep Learning: Unsupervised techniques as an optimization tool
- Communications Theory
- Information Theory
- Digital Communications
- Probability and Stochastic Processes
- Statistical signal processing: Estimation and Detection

Journal Publications

1. **J. K. Dassanayake**, R. Kulathunga, G. A. A. Baduge and M. Vaezi, "Unsupervised Learning-Based ISAC Waveforms," in *IEEE Wireless Communications Letters*, vol. 14, no. 9, pp. 2663-2667, Sept. 2025.
2. G. A. Baduge, M. Vaezi, **J. K. Dassanayake**, M. Z. Hameed, E. Ollila, and S. A. Vorobyov, "Frequency Range 3 for ISAC in 6G: Potentials and Challenges," in *IEEE Communications Magazine*, Mar. 2025 (accepted).

3. **J. K. Dassanayake**, D. Gunasinghe and G. A. A. Baduge, "Performance of SWIPT for IRS-Aided Massive MIMO With Statistical CSI-Based Phase-Shift and Transmit Power Optimization," in *IEEE Transactions on Vehicular Technology*, vol. 74, no. 4, pp. 5814-5832, April 2025.
4. **J. K. Dassanayake**, M Wang, MZ Hameed and N Yang, "Multi-Agent Deep-Q Network-Based Cache Replacement Policy for Content Delivery Networks," in *Future Internet, MDPI*, vol. 16, no. 8, Aug. 2024.
5. **J. K. Dassanayake**, D. Gunasinghe and G. A. Baduge, "Secrecy Rate Analysis and Active Pilot Attack Detection in IRS-Aided Massive MIMO Systems," in *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 2664-2679, 2024.
6. **J. K. Dassanayake**, D. Gunasinghe and G. A. Baduge, "ISAC Waveforms for ELAA-Based Ultra MIMO with Near-Field Spatially Correlated Fading," in *IEEE Transactions on Communications*, 2026. (Submitted, Manuscript ID: TCOM-TPS-26-1146).
7. Amirhossein Keshavarzchafjiri, **J. K. Dassanayake**, G. A. Baduge, and Mojtaba Vaezi "Gamma-Distributed Geometric Constellation for ISAC: Design and Analysis," in *IEEE Transactions on Wireless Communications*, 2026. (Submitted, Manuscript ID: Paper-TW-Apr-26-1324).

Conference Publications

1. **J. K. Dassanayake**, R. Kulathunga, and G. Amarasuriya, "Beamformer designs for swarm of repeater-aided massive MIMO ISAC" in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, May 2026 (accepted).
2. **J. K. Dassanayake**, T. Sakalasuriya, and G. Amarasuriya, "ISAC-Enabled SWIPT Through Massive MIMO" in *IEEE Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, 2026.
3. **J. K. Dassanayake**, T. Sakalasuriya, and G. Amarasuriya, "Secrecy Rate Analysis of RIS-Aided Massive MIMO with Active Adversaries" in *IEEE Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, 2026.
4. **J. K. Dassanayake** and G. Amarasuriya, "Near-Field Cell-Free Massive MIMO-Aided ISAC" in *IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, 2025.
5. **J. K. Dassanayake**, D. Gunasinghe and G. Amarasuriya, "Near-Field Performance of ELAA-Based ISAC" in *IEEE International Conference on Communication (ICC)*, Montréal, Canada, 2025.
6. R. Kulathunga, **J. K. Dassanayake**, and G. Amarasuriya, "Cell-Free Massive MIMO-Aided ISAC" in *IEEE International Conference on Communication (ICC)*, Montréal, Canada, 2025.
7. R. Kulathunga, **J. K. Dassanayake**, and G. Amarasuriya, "IRS-Aided Massive MIMO ISAC Systems" in *IEEE International Conference on Communication (ICC)*, Montréal, Canada, 2025.
8. M. A. Jayasinghe, **J. K. Dassanayake**, and G. Amarasuriya, "Trade-off Between Probability of Detection and Achievable Rate in Near-Field ISAC Systems" in *IEEE International Conference on Communication (ICC)*, Montréal, Canada, 2025.
9. **J. K. Dassanayake**, D. Gunasinghe and G. Amarasuriya, "Statistical CSI-Based IRS-Aided Massive MIMO SWIPT Systems," in *IEEE Global Communications Conference (GLOBECOM)*, Cape Town, South Africa, 2024.
10. **Dassanayake, D. J. K.**, et al. "Controller for the Power Injection from a Grid-Interfaced Energy Storage within an AC Micro Grid." *Smart Technologies for Energy, Environment and Sustainable Development*, Vol 1. Springer, Singapore, pp. 337-355. 2022.
11. **J. K. Dassanayake** and Gayan Amarasuriya "Secrecy Rate Analysis of IRS-Aided Massive MIMO with Active Pilot Contamination Attacks" in *IEEE International Conference on Communication (ICC)*, Rome, Italy, 2023 (accepted).

12. **J. K. Dassanayake**, R. Kulathunga, and Gayan Amarasuriya “Repeater-Aided Massive MIMO ISAC Systems ” in IEEE 104th Vehicular Technology Conference (VTC), 2026. (submitted).
13. Muhammad Zia Hameed, **J. K. Dassanayake**, and Gayan Amarasuriya “Inception-based Unified ISAC Waveform Learning” in IEEE 104th Vehicular Technology Conference (VTC), 2026. (submitted).

Book Chapters

1. **Dassanayake, D. J. K.**, et al. “Day-Ahead Solar Power Forecasting Using Artificial Neural Network with Outlier Detection.” Artificial Intelligence, Internet of Things (IoT) and Smart Materials for Energy Applications, CRC Press, 51-64, 2021.

Awards and Honors

- **Best Paper Award** for the work “ISAC-Enabled SWIPT Through Massive MIMO” in IEEE Computing and Communication Workshop and Conference (CCWC), Las Vegas, 2026.
- Awarded with the Southern Illinois University **Doctoral Fellowship** for the years 2023 - 2024 and 2024 – 2025.
- Awarded with the National Science Foundation (**NSF**) travel grant for conference travel (IEEE GLOBECOM 2025).
- Awarded with the National Science Foundation (**NSF**) travel grant for conference travel (IEEE ICC 2023).
- **Best Paper Award** for the research paper “Controller for the Power Injection from a Grid Interfaced Energy Storage within an AC Micro-Grid” in ICSTEESD-2020. This paper was based on my undergraduate project.

Research Experience

- **Optimal waveform/beamformer designs for integrated sensing and communication (ISAC)** (2024 - Present)
Extensive performance analysis of ISAC system models incorporated with extremely large antenna arrays, near-field propagation, cell-free architecture, RISs, and repeaters has been carried out. I have investigated joint beamforming, waveform designs, detectors, and Cramér-Rao bounds pertaining to such ISAC systems.
- **Learning-based waveform designs for ISAC** (2024 - Present)
This research explores learning-based waveform designs for ISAC to optimize trade-offs between communication and sensing performance. Deep learning techniques are leveraged to adaptively design waveforms that enhance spectral efficiency, target detection accuracy, and interference management. I focus on developing data-driven models to improve waveform adaptability in dynamic wireless environments, enabling robust and efficient ISAC implementations for next-generation networks.
- **Design of intelligent reflective surfaces (IRS) aided simultaneous wireless information and power transfer (SWIPT) massive multiple-input multiple-output (MIMO) systems** (2023)
Radio frequency signals can carry energy as well as information simultaneously. This leads to the study of SWIPT-based communication systems. The objective is to design systems that can harvest energy for low-power devices while transmitting information. I’ve done extensive research on such systems and formulated different optimization strategies to maximize the harvested energy and information rate in the systems.
- **Design of secure IRS-aided massive MIMO communication system contaminated with active eavesdroppers** (2022-2023)
Eavesdroppers are adversaries present in the system who intercept the confidential information of legitimate users. They contaminate the uplink channel by transmitting pilot signals similar to the users. This is known as active pilot contamination. The detection of these active pilot attacks is critical for secure communication. I’ve researched on the performance of such communication systems by analyzing the performance metrics and also designed two detectors based on the Neyman-Pearson and generalized likelihood ratio test criteria.

- **Efficient cache replacement algorithm for content delivery network (CDN) based on reinforcement learning** (2023)

This project was completed to fulfill a requirement for the course “Computer Network Systems Architecture”. A deep Q-learning-based reinforcement algorithm was employed to develop an efficient cache replacement policy for a CDN.

- **Design of a protection coordination scheme for a 22-Bus network** (2023)

This project was completed to fulfill a requirement for the course “Electric Power Distribution”. ETAP software was utilized in simulating the 22-bus network and protection coordination scheme was developed to ensure the system safety during permanent faults.

- **Undergraduate Project** (2019-2020)

The main focus of the study was to develop an innovative power flow controller for a grid-connected 1 kW three phase three level neutral point clamped multilevel inverter embedded with energy storage in an AC microgrid. The multilevel inverter was used to control active and reactive power flow to the microgrid. The power controller was designed based on synchronous reference frame theory and simulated using MATLAB® SIMULINK environment. Texas Instrument C2000™ F2879D Microcontroller was used for the real-time implementation of the controller.

Conferences and Experiences

- 2026 Attended IEEE Communication and Computing Workshop and Conference (CCWC) 2026 in **Las Vegas, Nevada, USA**.
- 2025 Attended IEEE Global Communications Conference (GLOBECOM) 2025 in **Taipei, Taiwan**.
- 2025 Mini workshop on “AI-aided ISAC” 2025 in **Southern Illinois University, Carbondale, USA** (organization committee member/presenter).
- 2025 Attended IEEE International Conference on Communication (ICC) 2025 in **Montreal, Canada**.
- 2024 Attended IEEE Global Communications Conference (GLOBECOM) 2024 in **Cape Town, South Africa**.

Technical Paper Presentations

- Presented the paper “**ISAC-Enabled SWIPT Through Massive MIMO**” at IEEE CCWC 2026 in Las Vegas, USA.
- Presented the paper “**Secrecy Rate Analysis of RIS-Aided Massive MIMO with Active Adversaries**” at IEEE CCWC 2026 in Las Vegas, USA.
- Presented the paper “**Near-Field Cell-Free Massive MIMO-Aided ISAC**” at IEEE ICC 2025 in Taipei, Taiwan.
- Presented the paper “**Near-Field Performance of ELAA-Based ISAC**” at IEEE ICC 2025 in Montréal, Canada.
- Presented the paper “**Statistical CSI-Based IRS-Aided Massive MIMO SWIPT Systems**” at IEEE GLOBECOM 2024 in Cape Town, South Africa.
- Presented the paper “**Impact of Imperfect SIC on Achievable Rates of RIS-Aided Massive MIMO With RSMA**” at IEEE GLOBECOM 2024 in Cape Town, South Africa.

Professional Services

- Technical Program Committee (TPC) Member in **International Communication Conference 2026**.
- Reviewer at **IEEE Transactions on Communications**.
- Reviewer at **IEEE Wireless Communications Letters**.
- Reviewer at **IEEE Communications Letters**.

- Reviewer at **IEEE Open Journal of the Communications Society**.
- Vice President of the Sri Lankan Student Association of Southern Illinois University Carbondale for the year 2023 - 2024.
- Served as a judge at Illinois Junior Academy of Science State Science Fair 2023.
- **Consultancy service provided for Sri Lanka Army - Artillery Division** (2021)
Member of the consultancy team who provided expert guidance on repairing a radar system. The system had power failures, communication equipment failures, and network reconfigurations.

Professional Experience

Aug. 2023 - July 2025	Graduate Research Assistant (Two consecutive Doctoral Fellowship Awards) , School of Electrical, Computer, and Biomedical Engineering, Southern Illinois University Carbondale, IL
Jan. 2019 - Apr. 2019	Electrical Engineer (Intern) , Ceylon Electricity Board — Thermal and hydro power plants, System control, Generation and transmission planning, Medium and low voltage distributions
Nov. 2017 - Feb. 2018	Electrical Engineer (Intern) , Airport and Aviation Services (Sri Lanka) Ltd. — Electrical maintenance, Electrical installations, Power Systems, Airfield lighting system

Teaching Experience

Jan. 2024 - Present	Graduate Teaching Assistant , School of Electrical, Computer, and Biomedical Engineering, Southern Illinois University Carbondale, IL — Providing assistance to advisor in teaching and grading (Primary courses: Principles of Communication, Introduction to Machine Learning, Probability and Statistics), Mentoring one undergraduate and graduate student
Jan. 2023 - May. 2023	Graduate Teaching Assistant , School of Electrical, Computer, and Biomedical Engineering, Southern Illinois University Carbondale, IL — Designing a robotics Lab Manual for undergraduate students.
Aug. 2022 - Dec. 2022	Graduate Teaching Assistant , School of Electrical, Computer, and Biomedical Engineering, Southern Illinois University Carbondale, IL — Course: Signals and Systems — Conducting lab sessions, grading quizzes, homework, and exams
Jan. 2022 - May 2022	Graduate Teaching Assistant , School of Electrical, Computer, and Biomedical Engineering, Southern Illinois University Carbondale, IL — Course: Computer Network Systems Architecture — Conducting lab sessions, grading quizzes, homework, and exams
Aug. 2020 - Oct. 2021	Instructor , Faculty of Engineering, University of Ruhuna, Sri Lanka — Conducting laboratory sessions, tutorials, and workshops for modules. Main focus has been on analog and digital electronics, wireless communication, control systems design, and power electronics.
Jan. 2019 - Dec. 2019	Student Instructor , Faculty of Engineering, University of Ruhuna, Sri Lanka — Conducting laboratory sessions.